

EDUCATION

Wuhan University | State Key Lab. LIESMARS

Wuhan, China | 2023.09 - 2026.06

M.Sc. in Cartography and Geographic Information Systems

- GPA: 3.55/4.00, Average Score: 89.40/100, IELTS: 7.0(6.0)
- Research area: Computer Vision and 3D Reconstruction

Wuhan University | School of Remote Sensing and Information Engineering

Wuhan, China | 2019.09 - 2023.06

B.Eng. in Remote Sensing Science and Technology

- GPA: 3.75/4.00, Average Score: 88.90/100, Recommended Exemption

PUBLICATIONS

1. **Xu Pan**, Zhenglin Wan, Xingrui Yu, Xianwei Zheng, Youkai Ke, Ming Sun, Rui Wang, Ziwei Wang, Ivor Tsang. **SA-VLA: Spatially-Aware Reinforcement Learning for Flow-Matching Vision-Language-Action Models**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2026. Available at <https://arxiv.org/abs/2602.00743>.
2. **Xu Pan**, Zimin Xia, Xianwei Zheng. **Scale-aware Co-visible Region Detection for Image Matching**. *ISPRS Journal of Photogrammetry and Remote Sensing [JCR Q1, IF 12.2]*, 2025. <https://doi.org/10.1016/j.isprsjprs.2025.08.015>
3. Ming Sun, Rui Wang, Xingrui Yu, Lihua Jing, Hangyu Du, Zhenglin Wan, **Xu Pan**, Ivor Tsang. **Towards Backdoor-Based Ownership Verification for Vision-Language-Action Models**. (*Under Review*), 2026.
4. **Xu Pan**, Qiyuan Ma, Mingyue Dong, He Chen, Wei Ji, Xianwei Zheng. **SAMatcher: Co-Visibility Modeling with Segment Anything for Robust Feature Matching**. *IEEE Transactions on Image Processing (Under Review)*, 2026.

EXPERIENCES

Nanyang Technological University (NTU)

Singapore | 2026.07 - present

Research Assistant | *School of Electrical and Electronic Engineering*

- Under the supervision of Prof. Ziwei Wang, at Perception and Embodied Intelligence (PINE) Lab

Agency for Science, Technology and Research (A*STAR)

Singapore | 2025.08 - 2026.06

Research Assistant | *Centre for Frontier AI Research (CFAR), Institute of High Performance Computing (IHPC)*

- Pioneered spatially-aware reinforcement learning adaptation for flow-based VLA models, preserving implicit spatial inductive bias through fused geometric prior and preventing degradation under distribution shifts.
- Designed geometry-aligned dense rewards and spatially-conditioned exploration, enabling stable RL fine-tuning, superior zero-shot spatial generalization, and robust transferable behaviors in complex manipulation scenes.

Baidu International Technology (Shenzhen) Co., Ltd.

Shenzhen, China | 2024.12 - 2025.03

Research Intern | *Foundation Public Technology Group, International Technology R&D Department*

- Developed DiT-based framework for controllable, arbitrarily long video generation via key-frame conditioning.
- Improved the visual fidelity of rendered text in image generation by designing specialized attention mechanisms.
- Contributed to the **SynClub** AI chat app through assessment of frontier multimodal generative models and optimization of text-to-image/video modules, delivering improved performance and higher-quality outputs.

PROJECTS

Intelligent Interpretation Platform for Remote Sensing Images Based on Baidu PaddlePaddle

2022.03 - 2022.08

 TwSphinx54/I2RSI

Web-based Global Natural Disaster Assessment System

2022.05 - 2022.09

 TwSphinx54/GNDAS

ACADEMIC ACTIVITIES

2026 Vision and Learning Seminar (Valse), China Society of Image and Graphics (CSIG) | Wuhan, China 2026.05.08 - 05.10

2025 Annual Conference on Photogrammetry and Remote Sensing, Chinese Society for Geodesy, Photogrammetry and Cartography (CSGPC) | Kunming, China 2025.07.28 - 07.30

AWARDS AND HONORS

- **Outstanding Graduating Graduate Student**, Wuhan University 2026.04
- **Outstanding Graduate Student**, Wuhan University 2025.11
- **Graduate Academic Excellence Scholarship**, Wuhan University 2025.11
- **Graduate Academic Excellence Scholarship**, Wuhan University 2024.11
- **Outstanding Student Leader**, Wuhan University 2024.10
- **Outstanding Student Club Leader**, Wuhan University 2024.08
- **Active Contributor to Social Activities**, School of Remote Sensing and Information Engineering 2023.05
- **Wuhan University Class C Scholarship**, Wuhan University 2022.10
- **National Second Prize**, China Software Cup College Student Software Design Competition 2022.08
- **Honorable Mention**, Mathematical Contest In Modeling 2022.03
- **Third Prize**, Asia and Pacific Mathematical Contest in Modeling 2021.12
- **First Prize in Hubei Division**, China Undergraduate Mathematical Contest in Modeling 2021.10
- **Outstanding Student**, Wuhan University 2021.09
- **Bronze Medal**, China Collegiate Algorithm Design & Programming Challenge Contest 2021.03
- **Second Prize in Final**, Translation & Interpreting Contest of Hubei Province 2020.05

SKILLS

Languages: Mandarin, English (IELTS 7.0)

Programming: Python, C++, MATLAB, HTML, CSS, JavaScript, C#, SQL, R

Soft Skills: Leadership, Project Management, Event Organization, Public Speaking, Design & Editing

CERTIFICATES

System Integration Project Management Engineer

Qualification Certificate of Computer and Software Technology Proficiency, MOHRSS & MIIT

Certificate of L4 in Network Engineer, L3 in Network Technology, L3 in Database Technology

National Computer Rank Examination, NEEA

ACADEMIC SERVICES

Member ISPRS Student Consortium (ISPRS SC)

Volunteer 2022/2023 International Graduate Workshop on Geo-Informatics
The 21st China Ocean Color Conference

LEADERSHIP

Chairman The 22nd LIESMARS Graduate Student Union

Director GeoScience Café Operation Center